

PVsyst - Simulation report

Grid-Connected System

Project: Battery (2)

Variant: 8.82

No 3D scene defined, no shadings

System power: 8.82 kWp

Kampung Chuang - Malaysia

Author

NORTHERN SOLAR SDN. BHD. (Malaysia)



Project: Battery (2)

Variant: 8.82

PVsyst V8.1.3

VC0, Simulation date:
with V8.1.3

NORTHERN SOLAR SDN. BHD. (Malaysia)

Project summary

Geographical Site

Kampung Chuang

Malaysia

Situation

Latitude **3.4869** °(N)

Longitude **101.6284** °(E)

Altitude 22 m

Time zone UTC+8

Project settings

Albedo 0.20

Weather data

Kampung Chuang

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Simulation for year no 1

No 3D scene defined, no shadings

Orientation #1

Fixed plane

Tilt/Azimuth **20 / 90** °

Orientation #2

Fixed plane

Tilt/Azimuth **20 / -90** °

Near Shadings

no Shadings

System information

PV Array

Nb. of modules

14 units

Pnom total

8.82 kWp

Inverters

Nb. of units

1 unit

Total power

5 kWac

Pnom ratio

1.76

User's needs

Unlimited load (grid)

Results summary

Simulation step

Hourly

Produced Energy

11544 kWh/year

Specific production

1309 kWh/kWp/year

Perf. Ratio PR

75.85 %

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Main results	5
Loss diagram	6
Predef. graphs	7
P50 - P90 evaluation	8



Project: Battery (2)

Variant: 8.82

PVsyst V8.1.3

VC0, Simulation date:
with V8.1.3

NORTHERN SOLAR SDN. BHD. (Malaysia)

General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 20 / 90 °

Orientation #2

Fixed plane

Tilt/Azimuth 20 / -90 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Average Height 2.6 °

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Astronergy

Model CHSM66RN(DG)/F-BH-630

(Custom parameters definition)

CHSM66RN(DG)F-BH-630.PAN

Unit Nom. Power 630 Wp

Array #1 - PV Array

Orientation 1

Tilt/Azimuth 20/90 °

Number of PV modules 7 units

Nominal (STC) 4410 Wp

Modules 1 unit x 7 In series

At operating cond. (50°C)

Pmpp 4115 Wp

U mpp 264 V

I mpp 16 A

Array #2 - Sub-array #2

Orientation 2

Tilt/Azimuth 20/-90 °

Number of PV modules 7 units

Nominal (STC) 4410 Wp

Modules 1 unit x 7 In series

At operating cond. (50°C)

Pmpp 4115 Wp

U mpp 264 V

I mpp 16 A

Total PV power

Nominal (STC) 9 kWp

Total 14 modules

Module area 37.8 m²

Cell area 35.3 m²

Inverter

Manufacturer Hoymiles

Model HIS-5L-G3

(Custom parameters definition)

Unit Nom. Power 5.00 kWac

Number of inverters 1 * MPPT 50% 0.5 unit

Total power 2.5 kWac

Operating voltage 90-460 V

Pnom ratio (DC:AC) 1.76

Number of inverters 1 * MPPT 50% 0.5 unit

Total power 2.5 kWac

Operating voltage 90-460 V

Pnom ratio (DC:AC) 1.76

Total inverter power

Total power 5 kWac

Number of inverters 1 unit

Pnom ratio 1.76

No power sharing

**Array losses****Array Soiling Losses**

Loss Fraction 1.5 %

Module Quality Loss

Loss Fraction -0.75 %

IAM loss factorASHRAE Param.: IAM = $1 - b_o (1/\cos i - 1)$ b_o Param. 0.05**Thermal Loss factor**

Module temperature according to irradiance

U_c (const) 20.0 W/m²KU_v (wind) 0.0 W/m²K/m/s**Module mismatch losses**

Loss Fraction 1.00 % at MPP

LID - Light Induced Degradation

Loss Fraction 1.45 %

Module average degradation

Year no 1

Loss factor 0.55 %/year

Imp / Vmp contributions 80% / 20%

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

Vmp RMS dispersion 0.4 %/year

DC wiring losses

Global wiring resistance 10 mΩ

Loss Fraction **2.00** % at STC**Array #1 - PV Array**Global array res. **550** mΩLoss Fraction **3.00** % at STC**Array #2 - Sub-array #2**Global array res. **275** mΩLoss Fraction **1.50** % at STC**System losses****Unavailability of the system**

Time fraction 1.0 %

3.7 days,

3 periods

AC wiring losses**Inv. output line up to injection point**Inverter voltage **220** Vac mono

Loss Fraction 1.00 % at STC

Global SystemWire section Copper **2 x 6** mm²Wires length **9** m



Project: Battery (2)

Variant: 8.82

PVsyst V8.1.3

VC0, Simulation date:
with V8.1.3

NORTHERN SOLAR SDN. BHD. (Malaysia)

Main results

Simulation step

Hourly

System Production

Produced Energy **11544 kWh/year**

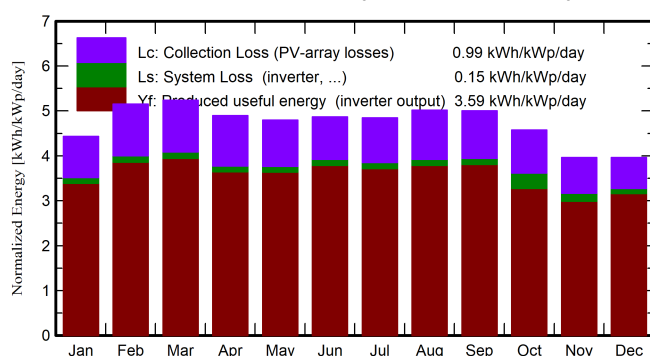
Specific production

1309 kWh/kWp/year

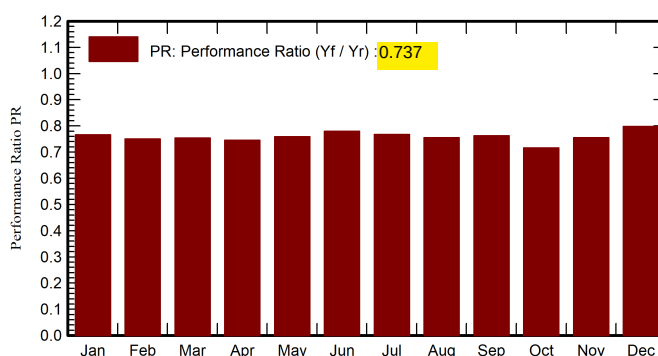
Perf. Ratio PR

75.85 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	PR ratio
January	141.8	70.84	26.85	137.5	130.6	962	928	0.766
February	149.0	72.94	27.46	144.4	137.8	989	955	0.750
March	167.7	87.92	27.68	162.4	155.2	1118	1079	0.754
April	151.7	74.54	27.73	146.8	140.1	1001	966	0.746
May	153.4	76.58	28.05	148.7	141.5	1031	996	0.759
June	150.5	73.96	27.96	146.0	138.9	1039	1003	0.779
July	155.4	74.21	27.56	150.4	143.4	1055	1018	0.767
August	160.8	80.04	27.59	155.7	148.6	1075	1037	0.756
September	154.9	82.48	27.18	150.1	143.1	1045	1009	0.762
October	146.2	81.34	27.07	141.9	134.8	991	896	0.716
November	122.9	70.64	26.68	118.9	112.6	840	792	0.755
December	127.2	79.83	26.64	123.0	116.5	897	866	0.798
Year	1781.5	925.32	27.37	1725.5	1643.0	12042	11544	0.758

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

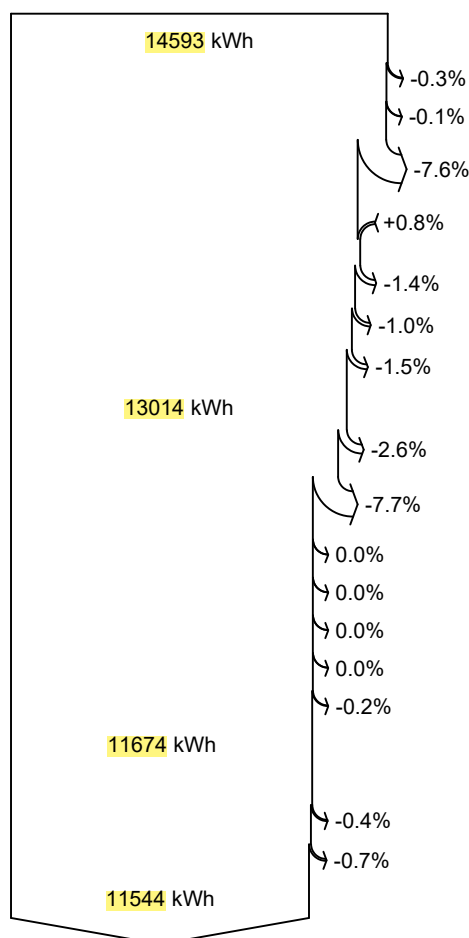
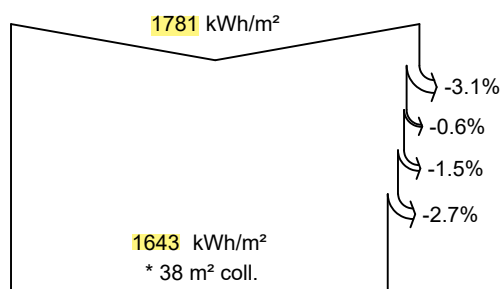
EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Far Shadings / Horizon

Soiling loss factor

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

Module Degradation Loss (for year #1)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

LID - Light induced degradation

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

AC ohmic loss

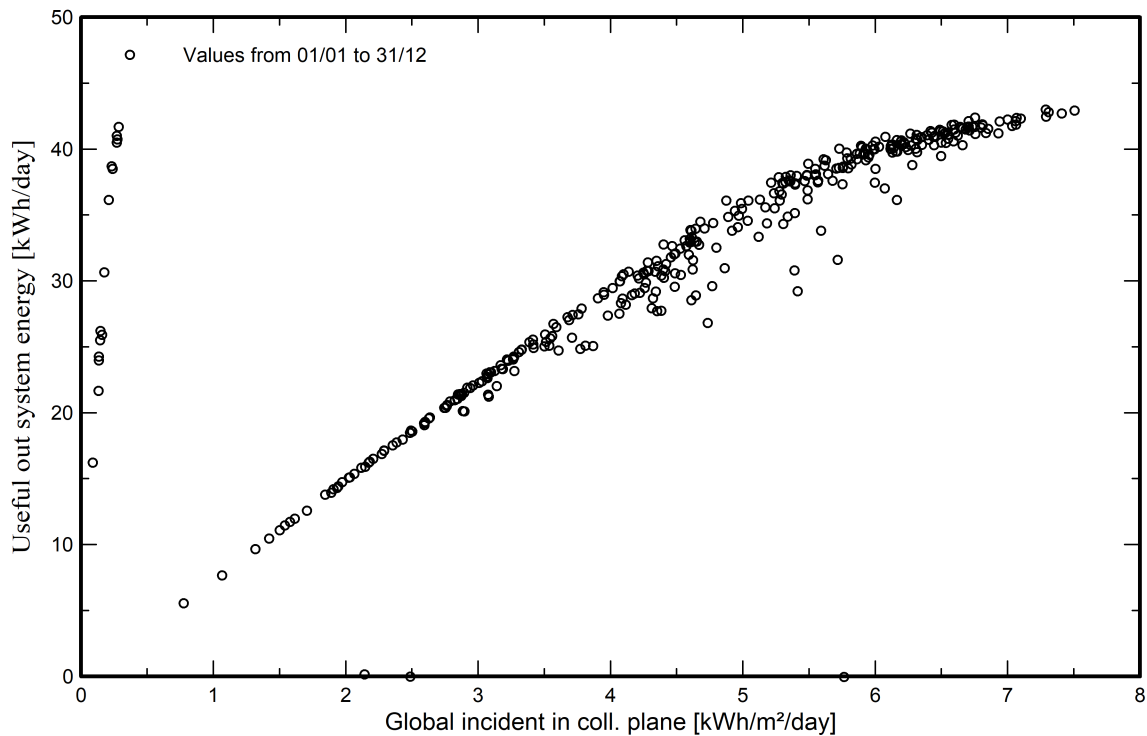
System unavailability

Energy injected into grid

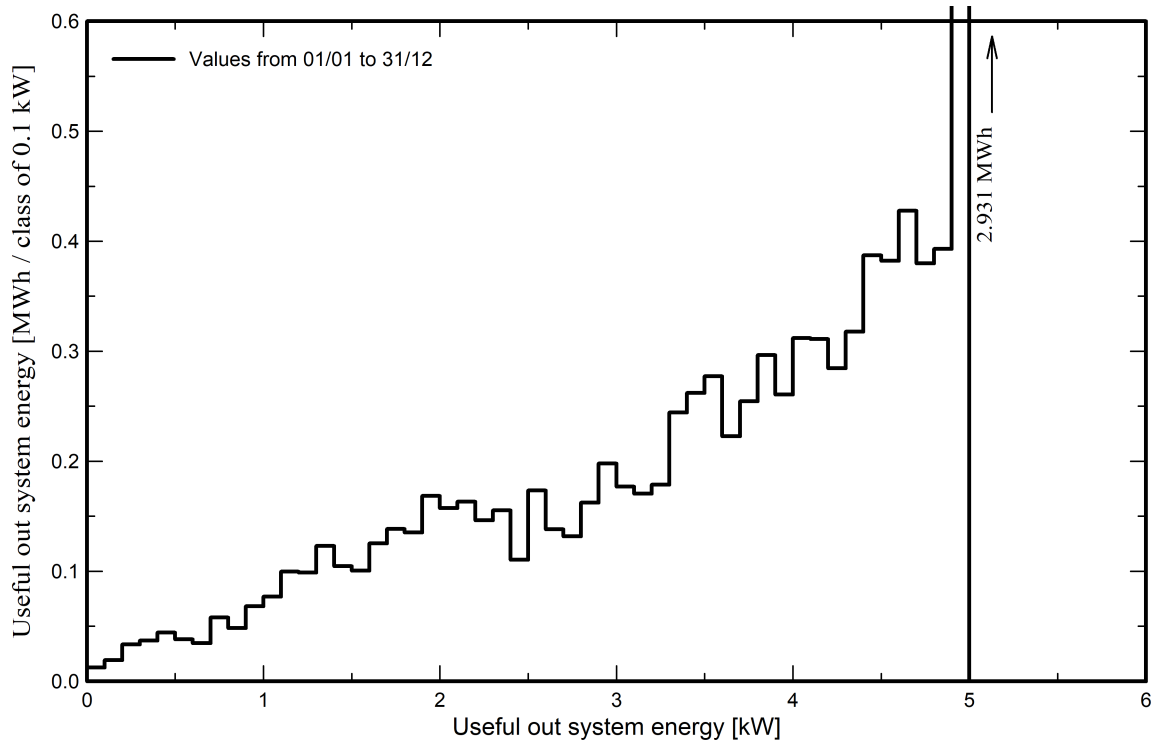


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution





P50 - P90 evaluation

Weather data

Source Meteonorm 9.0 dll, Sat=100%
Kind Specific year
Year Synthetic
Year-to-year variability(Variance) 4.0 %

Specified Deviation

Year deviation from average 0.0 %

Global variability (weather data + system)

Variability (Quadratic sum) 5.9 %

Simulation and parameters uncertainties

PV module modelling/parameters 1.0 %
Inverter efficiency uncertainty 0.5 %
Soiling and mismatch uncertainties 1.0 %
Degradation uncertainty 1.0 %
Custom variability 4.0 %

Annual production probability

Variability 685 kWh
P50 11544 kWh
P90 10665 kWh
P75 11082 kWh

Probability distribution

